Chaos-Based Multiscale Tensor Pyramid Network for Remote Sensing Scene Classification

Luobing Chen, Junjun Yin, *Senior Member, IEEE*, and Jian Yang, *Senior Member, IEEE*

***Abstract*—Remote sensing scene classification categorizes imaged areas by ground contents to provide semantic information for land use and land cover (LULC) monitoring. The remote sensing images exhibit diverse scene targets, textures, shapes, and spatial arrangements under varying solar illumination, therefore needing high-level feature analysis for effective scene characterization. Existing methods, including the widely used conventional neural networks, often fall into local minima during the optimization process due to the nonconvexity of the loss function over the parameter space. Furthermore, existing methods have two limitations in feature extraction: single-scale features are not adaptive to the spatial variation, and simply concatenating multiscale features neglects the geometric relationships among resolutions. Moreover, the flattening operation commonly used in network architectures tends to destroy the inherent topological structure of images, leading to a loss of spatial information in the scene representation. To address these issues, in this study, a Chaos-based multiscale tensor pyramid network (CMTPN) is proposed. First, a 3-D discrete chaotic system is proposed based on both the logistic map and the tent map to achieve a more uniform distribution and a broader full-chaotic map range. The map is then embedded into the neural network structure with the aim of obtaining an optimization result. Second, a multiscale tensor pyramid network is proposed. This network extracts local geometric details and global contextual information through parallel branches, in which the tensor contraction layer (TCL) is utilized to combine features from different scales by using attention mechanisms to adaptively integrate high- and low-resolution information, and meanwhile, the parallel tensor regression layer (TRL) breaks down the multiscale features using the Tucker decomposition for the final classifications. Effectiveness of the proposed algorithm is demonstrated through extensive experiments conducted on three remote sensing benchmark datasets: NWPU-RESISC45, Aerial Image Dataset (AID), and OPTIMAL-31, showing that the proposed CMTPN not only achieves superior classification accuracy but also maintains competitive computational efficiency, robustly outperforming state-of-the-art methods.**

***Index Terms*—Chaos map, remote sensing data classification, residual neural network, tensor decomposition.**

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I. Introduction

REMOTE sensing scene classification is an important task in remote sensing applications, offering essential informative support for geographic information systems (GISs) and Earth science research. However, as the volume of remote sensing data continues to grow, classification methods based on hand-crafted features show a series of limitations when dealing with complex-scene and large-scale data. These limitations include deficiencies in multispectral and hyperspectral information extraction, inefficient processing of extensive remote sensing data, and challenges in handling diverse feature categories. To address these issues, deep learning methods have emerged as a prominent approach in remote sensing data classification in recent years [1].

In recent years, the advancement of deep learning has brought neural networks back into wide attention. Since 2014, the field of remote sensing has gradually transitioned its focus to deep learning, and the related algorithms have made significant breakthroughs in many image analysis tasks, including land use and land cover (LULC) classification [2], scene classification [3], and target detection [4]. Yang et al. [5] proposed a deep learning framework for remote sensing image scene classification (RESISC). Their study demonstrated that incorporating multispectral bands (beyond RGB) enhances aerial scene classification accuracy. However, this approach encounters dimensionality challenges in high-dimensional feature representation. Anwer et al. [6] proposed to use an information-theoretic compression approach to address this issue. This method can obtain a compact set of deep color features without significant loss in accuracy. Yang et al. [7] proposed a new scene classifier named the graph scene classifier (GSCer). The advantage of this algorithm is that not only can the global semantics in high-resolution remote sensing images be fully explored, but the diverse hidden local semantics can also be extracted. Su et al. [8] proposed a new complete frequency channel attention (CFCA) network, which employs the low-frequency information of images, resulting in better classification and recognition capabilities. Duan et al. [9] developed a Swin Transformer with multiscale fusion (STMSF) and a spatial attention pyramid network (SAPN) to enhance multiscale feature representation in remote sensing imagery. Zhao et al. [10] proposed a local and long-range collaboration framework (L2RCF) to enhance the collaborative capabilities of convolutional neural networks and vision transformers. Wang et al. [11] proposed a multistage self-guided separation network (MGSNet) for remote sensing scene classification. This method combines the texture and morpho-

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logical features of images to guide feature learning, effectively reducing the impact of intraclass variation.

Although the previous research has achieved significant results in the field of remote sensing data classification, there are still challenges and limitations that require further consideration. First, artificial neural networks face limitations in research and applications due to their neglect of the dynamic characteristics inherent in biological neural systems. Artificial neural networks use the typical gradient descent algorithms for optimization, which can easily lead to local minima or infeasible solutions. Second, although significant efforts have been made to develop various networks to improve the accuracy, most approaches do not pay attention to preserving the spatial information when the flattening process is employed to transform the spatially structured data into a 1-D flattened representation. For the image scene classification task, the spatial arrangement in the original data is essential information. In addition, existing network structures often have defects in scale modeling. In single-scale paths, local details and global contextual information cannot be considered simultaneously, while simply stacked multiscale features fail to capture the geometric relationships across scales.

In the literature, some algorithms were proposed to address the issue of neural networks easily getting trapped in local minima. Wang [12] proposed using the Elitist firefly algorithm to optimize the structure of convolutional neural networks (CNNs). The global search strategy of the firefly algorithm can effectively avoid local minima, making it more effective for remote sensing image classification. Cheon [13] proposed combining the Kolmogorov–Arnold network (KAN) with the pretrained ConvNeXt architecture to improve performance in remote sensing image classification. This method reduces the risk of the network getting stuck in local minima through flexible activation functions and efficient parameter tuning. Although these studies have made some progress in addressing the local minimum problem and provided alternatives, most still rely on parameter fine-tuning and lack dynamic adjustment and global search capabilities, which limit their application and generalization in more complex scenarios.

In recent years, the discovery of chaos in the brain's cognitive processes has led to the understanding that chaos plays a pivotal role in the nervous system [14]. Chaos, as a prevalent nonlinear dynamic phenomenon, exhibits seemingly random motion but adheres to intricate internal laws. Neural networks utilize the ergodicity of chaotic dynamics to escape local minima during the search.

As an intrinsic representation of multidimensional data, tensors help to counteract structural collapse caused by vectorization, thereby preserving spatial hierarchies and reducing information loss. Through techniques such as tensor decomposition and convolution, tensor-based methods maintain structural features more effectively than vectorized approaches, leading to higher classification accuracy—especially in tasks requiring multidimensional spatial preservation. These methods excel with complex data and offer distinct advantages over traditional CNNs [15].

To overcome multiscale modeling limitations, pyramid architectures (e.g., feature pyramid networks) have become pivotal in remote sensing scene classification [16], [17]. These frameworks hierarchically extract and fuse features across spatial granularities—preserving fine details through high-resolution pathways while capturing contextual semantics via low-resolution branches—thereby resolving the single-scale constraints that impede joint optimization of local textures and global structures.

However, although tensor methods and multiscale pyramid networks demonstrate significant advantages in preserving spatial integrity and modeling scale diversity, the current research predominantly isolates these approaches. Investigations into synergistic integration—jointly preventing spatial structure destruction while enabling hierarchical cross-scale fusion—remain critically underexplored.

This isolation severely limits their effectiveness in remote sensing scene classification, where objects and structures exhibit vast scale variations and strong spatial dependencies. Models based solely on tensor operations maintain intricate spatial hierarchies yet often fail to integrate contextual cues across different scales—such as relating fine-grained textures in agricultural fields to broader landscape patterns. On the other hand, multiscale pyramid networks capture multilevel features but typically collapse structural details through pooling or vectorization, leading to the loss of critical geometric and topological information (e.g., field boundaries and building layouts).

As a result, the absence of a unified tensor-multiscale fusion mechanism often leads to misclassification of semantically complex scenes. Integrating these approaches is, therefore, essential for enhancing model discriminability in high-resolution Earth observation tasks.

In light of the above discussion, a multiscale tensor pyramid network based on a new 3-D chaos map is proposed for scene classification. The effectiveness of the algorithm is validated using NWPU-RESISC45, Aerial Image Dataset (AID), and OPTIMAL-31 remote sensing image datasets. In detail, our contributions to this article can be summarized as follows.

1) An ResNet network architecture based on a novel chaotic map is proposed. The chaotic map is able to realize almost the full map and has a wider chaotic range. With this feature, the model has a broader range of parameters during training, which is more helpful to improve the classification accuracy of the model.
2) This study proposes a novel multiscale tensor pyramid network. The tensor contraction layer (TCL) operates at the feature level: its compressed representation of global context serves as the input to an attention mechanism, which then dynamically fuses parallel high and low-resolution features. The tensor regression layer (TRL) operates at the prediction level: it replaces traditional fully connected layers by applying Tucker decomposition to multiscale features for low-rank compression, followed by an attention mechanism to fuse the final predictions. This dual innovation effectively preserves spatial information and achieves efficient modeling with minimal parameters.

The remaining sections of this article is organized as follows. In Section II, the proposed Chaos-based multiscale

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tensor pyramid network (CMTPN) architecture is described in detail, including the analysis of the new chaotic map and the introduction of the multiscale tensor pyramid network. In Section III, an experimental validation is carried out using three remote sensing image datasets, i.e., the NWPU-RESISC45 data, the AID data, and the OPTIMAL-31 data which are compared with the latest remote sensing image classification methods currently available. Finally, Section IV draws a conclusion.

II. Methodology

With the development of remote sensing technology, the volume of acquired remote sensing images is increasing, and the resolution is improving, which brings new challenges to image classification. Hand-crafted feature-based methods often perform poorly when dealing with complex scenes and high-dimensional data. Although deep learning, especially CNNs, usually achieves better performance in image classification, CNN methods still have some limitations in their practical applications. First, the nonconvexity of the CNN loss function, the high-dimensional complexity of the parameter space, and the limitations of gradient descent algorithms make CNNs prone to falling into local minima during optimization. Second, despite advancements, two fundamental limitations persist and are rarely addressed in unison: the loss of inherent geometric structure due to the destructive flattening operation in traditional layers [18], [19], [20], and the insufficient ability to capture and adaptively fuse multiscale features [21]. Most existing approaches tackle one of these issues in isolation—e.g., employing multiscale pyramids without preserving spatial relationships or utilizing tensor methods on single-scale features. This lack of a unified framework that simultaneously preserves spatial integrity and enables adaptive multiscale fusion ultimately hampers model performance on complex remote sensing scenes characterized by both rich details and large-scale variations. Therefore, in order to improve the accuracy and efficiency of remote sensing scene classification, we propose a new method of CMTPN. The proposed method hierarchically extracts and fuses multiscale tensor features under the pyramid structure, which not only better preserves the spatial information contained in the original data but also enhances the model's search capability through chaotic optimization strategies, thereby further improving the accuracy and efficiency of remote sensing scene classification [22].

A. Proposed Chatoic System

Chaos refers to the unpredictable and pseudorandom motion exhibited by deterministic dynamical systems due to their sensitivity to initial conditions. Chaos includes the following characteristics: long-term unpredictability, fractal properties, boundedness, and traversability. Chaotic motion is confined to a finite region, and the orbit never repeats [23]. The logistic map serves as a representative 1-D chaotic system and is expressed as follows:

$$x_{n+1} = \mu x_n (1 - x_n) \quad \mu \in [0, 4] \quad x \in [0, 1] \tag{1}$$

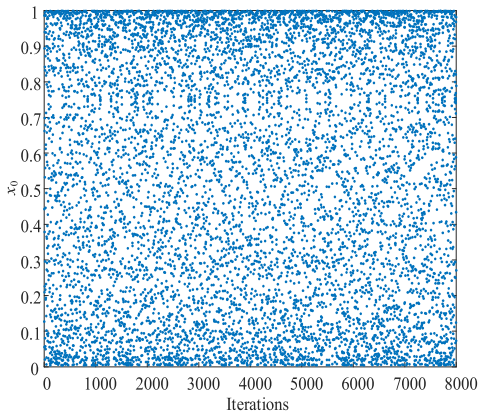


Fig. 1. Scatter plot of logistic chaotic map.

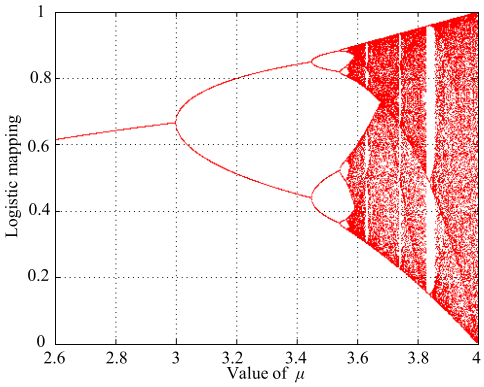


Fig. 2. Logistic chaos map bifurcation diagram.

where μ is a chaotic parameter. The scatter plot of the logistic map with $x_0 = 0.5$ is shown in Fig. 1. When μ takes a value of 3.98, the logistic map exhibits chaotic behavior. This implies that the generated sequence is nonperiodic, nonconvergent, and nondivergent. As shown in Fig. 1, the value of x is more likely to concentrate around 0 and 1 during the iteration process. Consequently, the logistic map has the disadvantage of uneven distribution [24].

The bifurcation diagram illustrates the process from bifurcation to chaos, and the analysis of chaotic characteristics is crucial [25]. When a large number of irregularly distributed points appear on the bifurcation diagram, it indicates that the system is chaotic. The bifurcation diagram for the logistic chaotic map is shown in Fig. 2. As the μ value continuously increases, more bifurcations occur in the system. In summary, when $3.5699456 < \mu \leq 4$, the results generated through iteration exhibit a pseudorandom distribution, dispersing across the entire range from 0 to 1.

The tent map is a function with parameter μ, which is a piecewise linear map. The definition is as follows:

$$x_{n+1} = \begin{cases} x_n/\mu, & 0 \leq x_n < \mu \\ (1 - x_n)/(1 - \mu), & \mu \leq x_n < 1 \end{cases} \tag{2}$$

where μ is the chaotic parameter. The scatter plot of the tent map with $x_0 = 0.5$ is shown in Fig. 3. When μ takes a value of 0.499, the tent map exhibits chaotic behavior.

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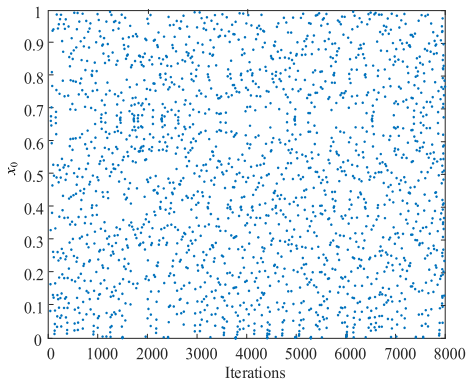


Fig. 3. Scatter plot of the tent chaotic map.

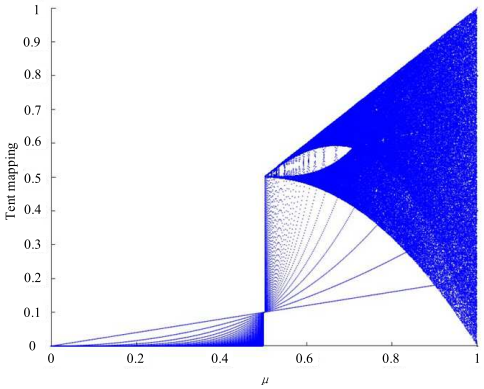


Fig. 4. Tent chaos map bifurcation diagram.

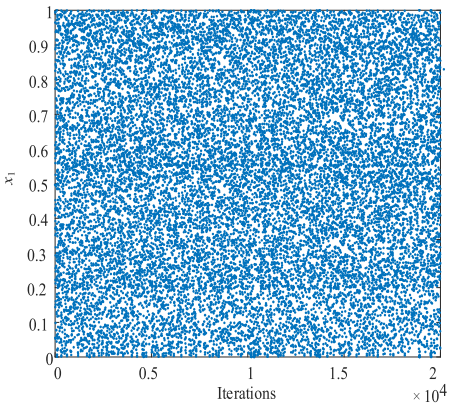


Fig. 5. New scatter plot of 3-D chaotic map.

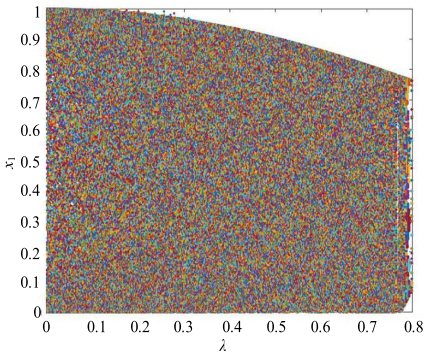


Fig. 6. x_1 with variation of λ bifurcation diagram.

The bifurcation diagram of the tent chaotic map is shown in Fig. 4, indicating that when $\mu \in (0, 1)$, the map system is in a chaotic state. Simultaneously, as the parameter μ approaches 1, the range of the map within the entire system is gradually extended to cover the full map. The iterative distribution of the tent map is notably uniform. However, the tent map cannot achieve a true full map interval and has a limited parameter range. When introducing it into neural networks, the improper parameter selection may not enhance the algorithm's performance and sometimes may lead to worse optimization results.

To overcome the above-mentioned disadvantages, we combine the logistic map with the tent chaotic map to create a new 3-D chaotic system. It possesses the intricate dynamic properties of the logistic chaotic system and the advantages of the tent chaotic system, including a uniformly distributed iteration, strong autocorrelation, and high applicability. Its definition is as follows:

$$\begin{cases} x1_{n+1} = \sin(\mu \cdot x1_n) \cdot \left(1 - \left(ax1_n^2 + bx2_n^2 + cx3_n^2\right)\right) \\ x2_{n+1} = \sin(\mu \cdot x2_n) \cdot \left(1 - \left(ax1_n^2 + bx2_n^2 + cx3_n^2\right)\right) \\ x3_{n+1} = \sin(\mu \cdot x3_n) \cdot \left(1 - \left(ax1_n^2 + bx2_n^2 + cx3_n^2\right)\right) \end{cases}$$
$$\begin{cases} xk_n = xk_n/\lambda, & 0 \le xk_n < \lambda \\ xk_n = (1 - xk_n)/(1 - \lambda), & \lambda \le xk_n \le 1 \end{cases} \tag{3}$$

where μ and λ are the chaotic parameters, $k = \{1, 2, 3\}$. Taking the initial values $(x_1, x_2, x_3) = (0.2, 0.4, 0.1)$, $\mu = 1.5$, and $\lambda = 0.499$, the scatter plot of the system is shown in Fig. 5. It indicates that the proposed system map is uniformly distributed in the full space between 0 and 1, demonstrating a stronger pseudorandom effect. This chaotic map can help to optimize neural network algorithms, enabling the model to exhibit chaotic behaviors such as uniform stochastic ergodicity and allowing the optimization process to break free from local minima, thereby enhancing the pattern recognition capability.

For the system shown in (3), set $\mu = 1.5$, $0 < \lambda < 1$, with initial values $(x_1, x_2, x_3) = (0.2, 0.4, 0.1)$, and parameters $a = 0.3, b = 0.4$, and $c = 0.3$. A bifurcation diagram is created with respect to the parameter λ in the λ–x plane, as shown in Fig. 6. Simultaneously, set $\lambda = 0.499$, $0 \le \mu \le 3$, with initial values (x_1, x_2, x_3) =(0.2, 0.4, 0.1), and parameters $a = 0.3, b = 0.4$, and $c = 0.3$. A bifurcation diagram is created with respect to the parameter μ in the μ–x plane, as depicted in Fig. 7. For application in remote sensing image recognition, the chaotic model is designed in a high-dimensional form.

When compared with the bifurcation diagrams of the foundational logistic map (see Fig. 2) and tent map (see Fig. 4), the advantages of the proposed system become evident. The logistic map operates chaotically only within a narrow parameter range ($\mu \approx [3.57, 4]$) and is plagued by a well-documented uneven distribution, with iterates clustering near the boundaries. While the tent map offers a more uniform distribution, its chaotic range [$\mu \in (0, 1)$] is still limited, and it inherently cannot achieve a full map range. In stark contrast, the proposed 3-D system demonstrates a significantly wider chaotic parameter range across both λ and μ. Furthermore, it

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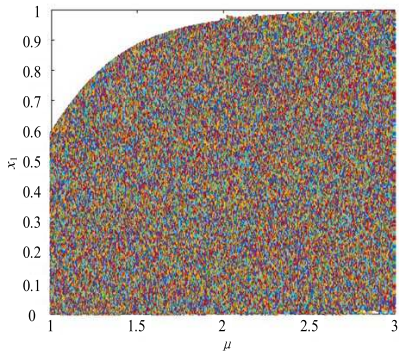


Fig. 7. x_1 with variation of μ bifurcation diagram.

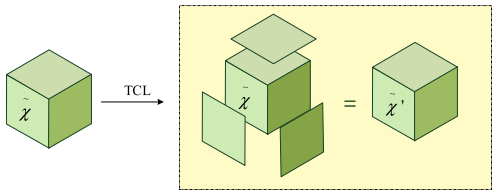


Fig. 8. TCL structural diagram.

provides a more uniform and comprehensive traversal of the entire [0,1] interval, as evidenced by the high upper bounds and lack of large gaps in its chaotic regions.

For the application in remote sensing image recognition, these properties are critically important. The chaotic model, designed in a high-dimensional form, benefits from an exceptionally wide chaotic parameter range. This translates to reduce sensitivity to the precise initial choice of parameters during the neural network training process, offering a broader and more robust operational regime. The ability to generate outputs that densely and almost uniformly traverse the state space enhances the stochastic search capability during optimization. This facilitates a more thorough exploration of the parameter space, which is more conducive to escaping local minima and ultimately improving the classification accuracy of the model.

B. Multiscale Tensor Pyramid Networks

1) Tensor Contraction Layers: TCLs are natural ways to introduce tensor operations into neural networks, achieving low-dimensional representations by contracting activation tensors. Compared with using fully connected layers for similar dimensionality reduction, TCL requires fewer parameters and computations while preserving the multilinear structure of the activation tensors. The network architecture is shown in Fig. 8.

Assume the input activation tensor $\mathcal{X}$, derived from the ResNet, has a size of $(S_0, I_0, I_1, \ldots, I_N)$. By using TCL, a smaller core tensor $\mathcal{X}'$ with a size of $(S_0, R_0, R_1, \ldots, R_N)$ is generated, defined by the following equation:

$$\mathcal{X}' = \mathcal{X} \times_1 V^{(0)} \times_2 V^{(1)} \times \ldots \times_{N+1} V^{(N)} \tag{4}$$

where $V(k) \in \mathbb{R}^{R_k \times I_k}$ is the projection matrix, $k \in \{0, \ldots, N\}$. The projection factor $V(k)$ is learned end-to-end through gradient backpropagation along with other network parameters.

Compared with fully connected layers, TCL requires significantly fewer parameters. For example, TCL has $\sum_{k=0}^{N} I_k \times R_k$ parameters, whereas fully connected layer has $\Pi_{k=0}^{N} I_k \times R_k$ parameters.

2) Tensor Regression Layers: In ResNet, it is common to generate predicted outputs by flattening activation tensors or applying spatial pooling operations. While these networks incorporate skip connections to alleviate the problem of vanishing gradients and improve training depth, flattening and pooling operations can still lead to the loss of the multilinear structure of activation tensors, leading to the information loss. TRLs generate outputs through low-rank tensor regression, which preserves the multimodal structure of activation tensors, thereby improving the accuracy of classification and regression tasks. The TRL structure diagram is shown in Fig. 9.

Assuming the input activation tensor $\mathcal{X}$, derived from the ResNet, has the size $\mathbb{R}^{S \times I_0 \times I_1 \times \cdots \times I_N}$, the output labels Y have the size $\mathbb{R}^{S \times O}$, and the regression weight tensor $\mathcal{W}$ has the size $\mathbb{R}^{I_0 \times I_1 \times \cdots \times I_N \times O}$, with $\mathcal{W}$ expressed using low-rank decomposition. The specific equation is as follows:

$$Y = < \mathcal{X}, \mathcal{W} >_N + b \tag{5}$$

where $< \mathcal{X}, \mathcal{W} >_N$ denotes the inner product of $\mathcal{X}$ and $\mathcal{W}$ along the N-mode, and b is the bias term. Through low-rank decomposition, the regression weight tensor $\mathcal{W}$ is represented as $\mathcal{W} = [G; U^{(0)}, \ldots, U^{(N)}, U^{(N+1)}]$, where G is the core tensor, and U is the factor matrices. This representation not only reduces the number of parameters but also retains multimodal information, enhancing the model's performance.

Through this decomposition method, TRL is able to reduce parameters and lower computational costs while retaining the multimodal structure of the input data, thereby improving the accuracy of classification and regression tasks.

3) Multiscale Tensor Pyramid Networks: Although TCL and TRL achieve feature dimensionality reduction and classification through tensor compression and regression, they have limitations in optical remote sensing image classification tasks. First, the current TCL only performs feature compression through single-branch convolution (nn. Conv2d), which makes it difficult to handle the significant size variations of multiscale targets in remote sensing images. As a result, the model struggles to acquire multigranularity information, leading to reduced classification accuracy for small targets [26]. Second, since TCL only acts on the deep outputs of ResNet, it does not integrate shallow features that contain rich edge and texture information, resulting in the model's difficulty in recovering spatial details lost by deep features. Finally, the model relies solely on single-scale features for classification, neglecting the complementary role of multiscale features, which can easily lead to misjudgment when processing large targets due to the local texture interference.

To address the above limitations, a multiscale tensor pyramid network is introduced. This model mainly consists of two parts: a fusion module and a regression module. The specific structure is shown in Fig. 10.

Let the input tensor be $\mathcal{X} \in \mathbb{R}^{B \times C \times H \times W}$, B is the batch size, C is the number of input channels, and H and W are the spatial resolutions. The high-resolution branch uses a standard convolutional layer (kernel size = 3 and padding = 1), batch normalization (BN), and rectified linear unit (ReLU) activation, thereby maintaining the input spatial dimensions.

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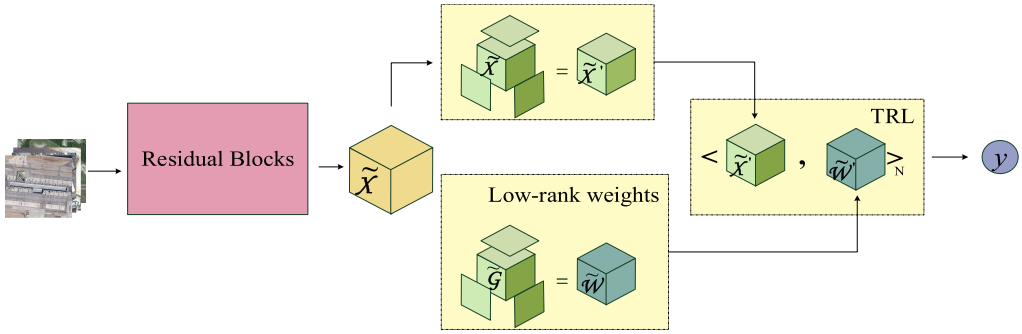


Fig. 9. TRL network structural diagram.

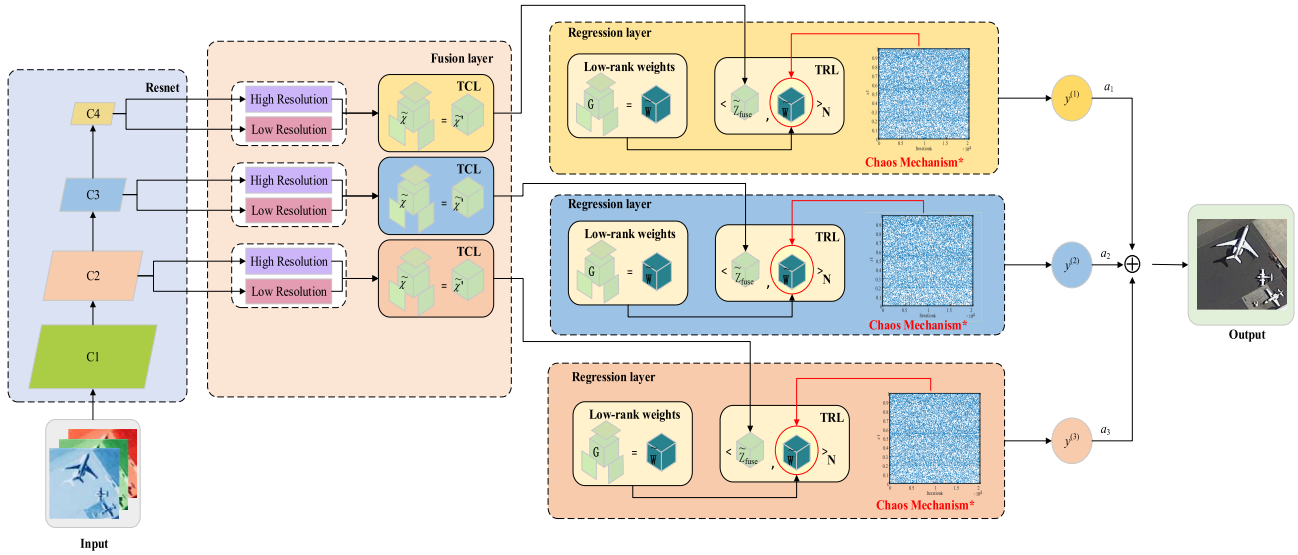


Fig. 10. Structural diagram of chaotic convolutional neural network based on tensor decomposition.

Mathematically, it can be expressed as follows:

$$x_{\text{high}} = \text{ReLU}\left(\text{BN}\left(\text{Conv2d}_{3\times3}(X)\right)\right) \in \mathbb{R}^{B\times C_{out}\times H\times W} \tag{6}$$

where $\text{Conv2d}_{3\times3}$ represents a convolutional operation with a kernel size of 3×3. BN is a normalization technique commonly used in deep neural networks [27]. During the training process of the network, the input distribution of each layer may change, a phenomenon known as internal covariate shift. BN reduces this shift by normalizing the inputs on each mini-batch of data, making the inputs of each layer have similar distributions, thereby accelerating the training process and improving model stability. Mathematically, the basic equation of BN can be written as follows:

$$\text{BN}(x) = \gamma \cdot \frac{x-\mu}{\sqrt{\sigma^2+\varepsilon}} + \beta. \tag{7}$$

First downsample to $H/2 \times W/2$ by 3×3 convolution (stride = 2) and do another 3×3 convolution, BN + ReLU. The expression is

$$x_{\text{low}} = \text{ReLU}\left(\text{BN}\left(\text{Conv2d}_{3\times3}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv2d}_{3\times3}(X)\right)\right)\right)\right)\right) \in \mathbb{R}^{B\times C_{out}\times\frac{H}{2}\times\frac{W}{2}}. \tag{8}$$

Next, to align with the high-resolution branch, the bilinear interpolation is applied to restore the resolution to (H, W)

$$x_{\text{low_up}} = \text{Interpolate}\left(x_{\text{low}}, size = (H, W)\right) \in \mathbb{R}^{B\times C_{\text{out}}\times H\times W}. \tag{9}$$

In the fusion module, the process of fusion using global statistical information and factor matrices is also included. It can be divided into the following steps.

The global average pooling is applied to the input of the same branch to obtain channel-level global vectors

$$v = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} X(b,:,i,j) \in \mathbb{R}^{B\times C}. \tag{10}$$

This step is equivalent to performing an n-mode product on the spatial dimensions or directly averaging over (i, j).

Next, $\mathcal{V}$ is projected to the high and low subspaces using two factorization matrices $F_{\text{high}}, F_{\text{low}} \in \mathbb{R}^{B\times C_{\text{out}}}$

$$h = v \cdot F_{\text{high}} \in \mathbb{R}^{B\times C_{\text{out}}}, l = v \cdot F_{\text{low}} \in \mathbb{R}^{B\times C_{\text{out}}} \tag{11}$$

where h and l are spliced along the channel dimension and then multiplied with a core tensor $\in \mathbb{R}^{(2C_{\text{out}})\times C_{\text{out}}}$

$$f = [h, l] \in \mathbb{R}^{B\times(2C_{\text{out}})}, f' = f \cdot \text{Core} \in \mathbb{R}^{B\times C_{\text{out}}}. \tag{12}$$

Therefore, f' is reshaped to a tensor of shape $[B, C_{\text{out}}, 1, 1]$, resulting in a tensor with C_{out} channels and spatial dimensions of 1×1

$$f'_{\text{reshape}} = \text{reshape}\left(f'\right) \in \mathbb{R}^{B\times C_{\text{out}}\times1\times1}. \tag{13}$$

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At this stage, an attention generator with a similar structure is employed

$$\begin{aligned}&\text{AttnGen}(z)\\&=\text{Soft max}\left(\text{Conv2d}_{1\times1}\left(\text{AdaptiveAvgPool2d}(1)(z)\right)\right)\\&\quad\in\mathbb{R}^{B\times2\times1\times1}\end{aligned}\tag{14}$$

where $z = f'_{\text{reshape}}$, AdaptiveAvgPool2d(1) is applied to an input tensor of shape $[B, C_{\text{out}}, 1, 1]$ without further dimensionality reduction, primarily to ensure compatibility with other potential input sizes. The final output of AttnGen(z) is an attention weight tensor of shape [B, 2, 1, 1], where the first channel corresponds to the high-resolution branch weight (denoted by α) and the second channel corresponds to the low-resolution branch weight (denoted by β).

After obtaining the attention weights α and β, broadcast them to match the shapes of x_{high} and $x_{\text{low_up}}$, and the elementwise multiplication is then performed

$$x' = \alpha \odot x_{\text{high}} + \beta \odot x_{\text{low_up}}\tag{15}$$

where $\alpha = \text{AttnGen}(f')_{:,0:1} \in \mathbb{R}^{B\times1\times1\times1}$ and $\beta = \text{AttnGen}(f')_{:,1:2} \in \mathbb{R}^{B\times1\times1\times1}$. This step does not alter the spatial resolution because α and β are broadcast across the spatial dimensions and multiplied with a tensor of shape $[B, C_{\text{out}}, H, W]$. As a result, the final output y retains its spatial dimensions (H, W).

After the fusion layer, the following features are obtained:

$$f^{(1)}\in\mathbb{R}^{B\times C_1\times H_1\times W_1}, f^{(2)}\in\mathbb{R}^{B\times C_2\times H_2\times W_2}, f^{(3)}\in\mathbb{R}^{B\times C_3\times H_3\times W_3}.$$

For the ith scale, after removing the batch dimension, the data can be represented as a 3-D tensor, denoted by $\mathcal{F}^{(i)} \in \mathbb{R}^{C_i\times H_i\times W_i}$. We, then, define three sets of factor matrices as follows:

$$U_C^{(i)} \in \mathbb{R}^{R_C\times C_i}, U_H^{(i)} \in \mathbb{R}^{R_h\times H_i}, U_W^{(i)} \in \mathbb{R}^{R_w\times W_i}.$$

The projection result of the ith scale can be written as follows:

$$\mathcal{Z}^{(i)} = \mathcal{F}^{(i)} \times_1 U_C^{(i)} \times_2 U_H^{(i)} \times_3 U_W^{(i)} \in \mathbb{R}^{R_c\times R_h\times R_w}.\tag{16}$$

For each scale i, a core tensor $\mathcal{G}$ is introduced with the following expression:

$$\mathcal{G}^{(i)} \in \mathbb{R}^{R_c\times R_h\times R_w\times\text{num_classes}}.\tag{17}$$

Then, the prediction of this scale is

$$\begin{aligned}y^{(i)} &=< \mathcal{Z}^{(i)}, \mathcal{G}^{(i)} >\\&=\sum_{c=1}^{R_c}\sum_{h=1}^{R_h}\sum_{w=1}^{R_w}\mathcal{Z}^{(i)}(c,h,w)\,\mathcal{G}^{(i)}(c,h,w,) \in \mathbb{R}^{\text{num_classes}}.\end{aligned}\tag{18}$$

For batch B, the shape of each scale output $y^{(i)}$ is $[B, \text{num_classes}]$. The outputs from all scales are stacked to obtain the following format:

$$Y \in \mathbb{R}^{B\times n\times\text{num_classes}}\tag{19}$$

where n is the number of scales. An attention module (a small MLT) is then used to compute the attention weights a_i for each scale, i.e.,

$$a = \text{soft max}\left(f_{attn}(Y)\right) \in \mathbb{R}^{B\times n\times1}.\tag{20}$$

Finally, it is fused y, i.e.,

$$y = \sum_{i=1}^{n} a_i y^{(i)} \in \mathbb{R}^{B\times\text{num_classes}}.\tag{21}$$

This allows the model to dynamically adjust the weights based on the importance of each scale to the task.

C. Proposed Algorithm

Based on the above discussion, we apply the chaos theory proposed in Section II-A to the above multiscale tensor pyramid network structure. Algorithm 1 and pseudocode are shown as follows.

Step 1: Input the high-dimensional activation tensor X and output labels Y. Apply a series of residual blocks to extract feature representations.

Step 2: Extract features from each residual module and group them into high-resolution and low-resolution branches. Then, apply a tensor decomposition technique to fuse the multibranch features and obtain the final output f_l.

Step 3: Input f_l into the TRL module for multiscale modeling and classification (see Section II-B3 for the structure and mathematical formulation of TRL).

Step 4: Randomly initialize the TRL weights W_0. Let W'_0 be the next iteration of W_0 calculated by the gradient descent method. Set $W' = W'_k$ and $k = 0$.

Step 5: Let $W_{k+1} = C(W'_k)$ be the initial value, and calculate W'_{k+1} using the gradient descent method, where $C(\cdot)$ is the chaotic theory proposed in Section II-A (3).

Step 6: If the loss function $L(W'_{k+1}) < L(W'_k)$, set $W' = W'_{k+1}$. Otherwise, W' remains unchanged.

Step 7: If W' remains unchanged after several iterations, the algorithm terminates. Otherwise, return to Step 5.

The structural diagram is shown in Fig. 10.

The ResNet part (blue module) primarily functions to extract features from the input image and enhance the training efficiency through a residual learning structure. The ResNet backbone consists of four convolutional modules ($C1$–$C4$), each responsible for processing features at different semantic levels. $C1$ is the initial convolutional layer in the network. It applies a 7 × 7 convolution with a stride of 2 to the input RGB image, generating a low-level feature map. This feature map primarily captures basic visual patterns, such as edges and textures. $C2$ is the first residual block in the network. It consists of multiple 3 × 3 convolutional layers and processes the feature maps from $C1$ to extract mid-level semantic information, such as object boundaries and basic shapes. $C3$ is the second residual block in the backbone. It deepens the representation through additional 3 × 3 convolutions and skip connections, capturing more complex visual patterns, including textures and intra-object boundaries. $C4$ is the final residual block in the

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Algorithm 1 CMTPNs

Input:
- High-dimensional input tensor $\mathcal{X}$ and corresponding label Y.
- ResNet backbone for feature extraction.
- MultiScale TCL module for high/low Resolution Branch.
- TRL for multi-scale classification.
- Chaos-related hyperparameters and optimization settings.

Output:
- Final classification predictions.

1: Feature Extraction and Intra-layer Fusion:
For each selected residual layer l (e.g., layer1, layer2, layer3):
a. Extract intermediate feature map F_ ℓ from the backbone.
b. Split F_ ℓ into two branches:
- High-resolution branch: F_high = HighRes(F_ ℓ)
- Low-resolution branch: F_low = LowRes(F_ ℓ)
c. Fuse the branches using MultiScaleTCL:
- f_ ℓ = MultiScaleTCL(F_ ℓ) (This module performs Tucker-based compression on the input via n-mode-like operations and applies attention-based fusion.).
if performing multi-scale only in the last layer then:
- Let f_final = f_last (i.e., only process layer3)
else:
- Gather fused features from all chosen layers: f1, f2, f3,... Concatenate these features along the channel dimension to form a unified multi-scale representation f_final.

2: Multi-scale Classification via TRL:
a. Optionally apply global pooling and dropout to f_final.
b. Feed f_final into the TRL module:
- Project the input features into a lower-rank space using learned factors.
- Perform rank-R factorization with a core tensor and projection factors.
- Compute per-scale logits and combine them using an attention mechanism.
c. Apply softmax to obtain predicted class probabilities.

3: Initialize weights & variables
a. Randomly initialize TRL weights (core & factors): W_0 = RandomInitializeWeights
b. Perform an initial gradient descent step: W'_0 = GradientDescent(W_0).
c. Set $W' \leftarrow W'_k$, and initialize iteration counter k = 0 and no_change_count = 0.

4: Chaos-Based Weight Optimization Iteration:
while no_change_count < N do:
a. Apply chaotic perturbation:
$Wk_{+1} \leftarrow$ ChaosOperation(W'_k)
b. Perform gradient descent:
$W'_{k+1} \leftarrow$ GradientDescent(W_{k+1})
c. Compare losses:
if loss(W'_{k+1}) $\leq$ loss(W'_k):
Set $W' \leftarrow W'_{k+1}$
else:
Keep W' unchanged
increment no_change_count by 1.
e. Increment iteration counter: k $\leftarrow$ k + 1.

5: Final Model Output and Testing:
a. Use the final optimized weights W' in the TRL.
b. Conduct testing on the test set using the final model.
c. Output final predictions for each sample.

End Algorithm.

network. It outputs a smaller sized high-level feature map that captures the most abstract representations.

Critically, while each module ($C2$–$C4$) operates at a distinct semantic level, the features within a single level still contain multifaceted information. To fully exploit this without losing original details, our design introduces parallel high- and low-resolution processing paths at these key semantic levels.

The fusion layer (orange area) merges these feature maps from different spatial resolutions within the same semantic level. The high-resolution path preserves fine-grained spatial details (e.g., precise edges and textures) by directly using the output feature map from a module like $C2$. Conversely, the low-resolution path aims to capture richer contextual and global information by downsampling the same input feature map (e.g., from $C2$), which expands the receptive field of each feature point at the cost of some spatial detail. The core task of the TCL module is to adaptively fuse information from these two branches: it learns to prioritize detailed features from the high-resolution path for areas requiring precise localization, while leveraging contextual clues from the low-resolution path for regions requiring semantic classification.

Through the TCL module, the network effectively integrates these multiresolution features and enhances its ability to capture multiscale information. After integrating the outputs of each TCL module, the network performs dimensionality reduction and feeds the result into the regression layer for final prediction.

In summary, the network processes information along two dimensions: vertically through semantic levels ($C1$–$C4$ of decreasing spatial resolution and increasing abstraction), and horizontally by maintaining and fusing multiple spatial resolutions within key semantic levels. This design equips the model with both a "magnifying glass" (high-resolution path) for detail and a "wide-angle lens" (low-resolution path) for context at each pivotal stage, enabling more accurate and robust comprehension of complex scenes.

The regression layer is responsible for mapping the fused feature maps to the final categories. In contrast to conventional flattening followed by fully connected layers, TRL applies a multilinear low-rank decomposition to project the input features. This design not only reduces the number of parameters but also preserves the spatial structure of the input features by avoiding destructive flattening. The outputs from all fused branches are processed jointly to generate the final classification results, ensuring comprehensive utilization of multiscale features.

After completing the initial training of the regression layer, the key parameters of TRL (core tensor and factor matrix) are further optimized using a chaotic control mechanism to achieve better convergence.

Chaos has the characteristics of ergodicity and initial sensitivity. Applying the chaos theory to neural networks can effectively avoid the local minimum problem. In addition, it achieves more stable and improved optimization results. Chaotic neural networks adopt the advantages of discrete chaos to solve the local minimum problem. The values of chaos at the adjacent moments are discrete, which means that they are not in a small neighborhood. The search process can jump out of local neighborhoods to solve the problem of local minimization. Chaos also gives neural networks higher search

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Fig. 11. RESISC45 dataset.

efficiency, generalization ability, and pattern recognition ability than conventional neural networks (CNNs).

The synergistic integration of chaotic optimization with our multiscale tensor pyramid network creates a powerful framework that leverages the strengths of each component: the multiscale pyramid ensures comprehensive feature extraction across spatial granularities, the tensor operations preserve critical structural information throughout the processing pipeline, and the chaotic dynamics enable effective escape from local optima during the optimization process. This combined approach provides a new direction for developing robust neural network techniques capable of handling the complex challenges presented by the modern remote sensing data.

III. Experiments

To ensure the reproducibility and reliability of the experimental results, we first describe the computational environment used for all experiments. All models were implemented in PyTorch and executed on a workstation equipped with an Intel Xeon Gold 6342 CPU (2.80 GHz) and an NVIDIA A800 80-GB PCIe GPU.

To evaluate the effectiveness of the proposed chaotic optimization-based scene classification method, we conducted experiments on the NWPU-RESISC45 dataset, the AID, and the OPTIMAL-31 dataset for ablation studies and comparisons with state-of-the-art methods. These datasets contain strong spatial structural information, which can be neglected by convolutional neural networks when processing high-dimensional inputs with complex backgrounds. Furthermore, they pose two major challenges: high interclass similarity and significant intraclass variation due to differences in viewing angles and scales. These characteristics reduce the robustness of traditional neural networks when dealing with such diverse and complex image scenes [28].

The RESISC45 dataset [29], designed for RESISC, contains 31 500 RGB images (256 × 256 pixels), distributed across 45 scene classes with 700 images per class. This dataset is characterized by its large scale, variety, high intraclass diversity, and interclass similarity, making it less prone to accuracy saturation. The intraclass diversity in RESISC45 allows us to test the chaotic optimization module's ability to handle feature variations. Fig. 11 illustrates an example.

AID [30] is a large-scale remote sensing image classification dataset released by researchers at Wuhan University.



Fig. 12. AID dataset.



Fig. 13. OPTIMAL-31 dataset.

It contains 10 000 high-resolution images (600 × 600 pixels) acquired from Google Earth. This dataset includes 30 categories, such as airports, forests, and residential areas. Each class corresponds to a specific geographic or man-made scene type. All image labels were manually and accurately annotated. The AID dataset is widely used in remote sensing image classification, feature extraction, representation learning, and cross-domain adaptation studies. This dataset helps to evaluate and test the performance of machine learning and deep learning models in RESISC tasks. Fig. 12 illustrates an example.

The OPTIMAL-31 dataset [31] is a high-resolution RESISC dataset, released by Northwestern Polytechnical University in 2019. It includes 1860 images across 31 scene classes, with 60 images per class. Each image has a resolution of 256 × 256 pixels and is sourced from Google Earth, encompassing various scene types such as urban areas, forests, beaches, and more. Fig. 13 illustrates an example.

A. Validation Using the RESISC45 Dataset

The dataset was randomly split using stratified sampling to preserve the original class distribution. Specifically, 50% of the samples were used for training, and 50% for testing. The model was trained using the AdamW optimizer with a learning rate of 0.0001 and a weight decay of 8×10^{-4}. The learning rate followed a cosine annealing schedule over 70 epochs, with a batch size of 32 for training. To address class imbalance, the focal loss was employed (modulation factor $\alpha = 0.25$ and focusing parameter $\gamma = 2.0$) [32]. In addition, label smoothing

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TABLE I
ABLATION EXPERIMENT OF THE RESISC45 DATASET

Algorithm	Accuracy	Kappa value	Recall rate	F1 value	Precision
ResNet-50	84.6%	0.839	84.6%	84.6%	86.6%
C-ResNet-50	90.0%	0.907	91.0%	90.9%	90.4%
T-ResNet-50	89.3%	0.887	88.9%	88.7%	89.8%
LT-ResNet-50	92.0%	0.929	92.8%	92.8%	91.9%
TT-ResNet-50	91.8%	0.925	92.3%	92.5%	91.8%
T-Chaotic ResNet-50	95.9%	0.957	95.9%	95.9%	95.9%
Proposed algorithm	**98.2%**	**0.981**	**98.2%**	**98.2%**	**98.9%**

($\varepsilon = 0.1$) was used to improve generalization. Early stopping was employed with a patience threshold of 40 epochs. To balance exploration and convergence stability, the parameters in (3) for chaotic weight optimization were set to $a = 0.3$, $b = 0.4$, $c = 0.3$, $\mu = 1.5$, and $\lambda = 0.45$.

To evaluate the effectiveness of the proposed method, we conducted two sets of experiments: ablation studies and comparative analyses. The ablation study was conducted to assess the contribution of each component in the model. The evaluated variants included ResNet-50, tensor-based ResNet-50 (T-ResNet-50), ResNet-50 integrated with the newly designed chaotic mapping (C-ResNet-50), T-ResNet-50 integrated with a logistic map (LT-ResNet-50), with a tent map (TT-ResNet-50), and based on the proposed chaotic mapping and tensor ResNet (T-Chaotic ResNet-50). The final integrated model was also included for comparison.

To compare with state-of-the-art methods, both classical and deep learning-based approaches were considered. These included classifier + detector [5], multifeature fusion [6], GSCer [7], CFCA [8], STMSF [9], L2RCF [10], MGSNet [11], ShuffleNet CNN [33], VGG-19 [34], and snapshot ensemble-based ResNet (SnapEnsemResNet) [35]. To validate improvements in escaping local minima, comparisons were also made with enhanced metaheuristic algorithms, including the improved Firefly algorithm [12] and KAN [13]. A total of 20 independent experiments were conducted, and the average results were reported for evaluation.

1) Ablation Experiment on the RESISC45 Dataset: An ablation study was conducted on the RESISC45 dataset to comprehensively evaluate the effectiveness of each individual component within the proposed framework, and the results were summarized in Table I.

The standard ResNet-50 baseline achieves an overall accuracy of 84.6%, serving as a reference for architectural improvements. The C-ResNet-50 leads to a notable improvement, increasing accuracy to 90.0%. This highlights the potential of chaotic dynamics in mitigating local minima and enhancing generalization. Separately, the incorporation of T-ResNet-50 achieves 89.3% accuracy, demonstrating its effectiveness in capturing multidimensional spatial context.

Specifically, LT-ResNet-50 and TT-ResNet-50 achieve 92.0% and 91.8% accuracy, respectively, suggesting that traditional chaotic functions (e.g., logistic and tent maps) may lack sufficient expressive complexity to support further performance gains. In contrast, the proposed T-Chaotic ResNet-50, which combined the novel chaotic mapping with tensor modeling, achieving 95.9% accuracy and outperforming them across all metrics. This improvement demonstrates the strong synergy between the proposed chaotic mechanism and tensor structures.

Finally, the full version of the proposed algorithm, integrating the chaotic mapping, tensor decomposition, and multiscale pyramid structure, achieves the best performance overall: accuracy of 98.2%, Kappa of 0.981, recall of 98.2%, $F1$ score of 98.2%, and precision of 98.9%.

To visualize the distribution of high-dimensional features in low-dimensional space, t-distributed stochastic neighbor embedding (t-SNE) visualizations are presented in Fig. 14.

The t-SNE visualization results of the original ResNet are shown in Fig. 14(a). The scatter points appear relatively dispersed, and noticeable category overlap in the central region indicates limited discriminative capability. In contrast, Fig. 14(b) illustrates the t-SNE results of ResNet with tensor integration (T-ResNet), which exhibits improved category separation with reduced overlap, particularly in the edge regions where class boundaries become more apparent. Subsequently, Fig. 14(c) displays the visualization of the chaos-based ResNet (C-ResNet), where most categories form tighter clusters with clearer boundaries, although some scattered points remain in the peripheral areas. The combination of tensor modeling and chaotic mapping, as demonstrated in Fig. 14(d) (T-Chaotic ResNet), further enhances category clustering by significantly reducing point overlap and improving intercategory separability. The t-SNE visualization of the proposed method, i.e., the chaos-based multiscale tensor pyramid ResNet, is presented in Fig. 14(e). The proposed method achieves the most compact clustering, with clearly separated categories in both central and peripheral regions. These results confirm the superior category discrimination capability of the proposed method.

2) Comparative Experiment on the RESISC45 Dataset: The performance comparisons of various remote sensing image classification algorithms using the RESISC45 dataset are shown in Table II.

The proposed algorithm achieves accuracy of 98.2%, surpassing the previously reported highest value of 97.2% (L2RCF). The Kappa coefficient also reaches 0.981, indicating that the model has high consistency and stability in multiclass classification problems. Moreover, it achieves the highest precision (98.9%) among all compared methods, demonstrat-

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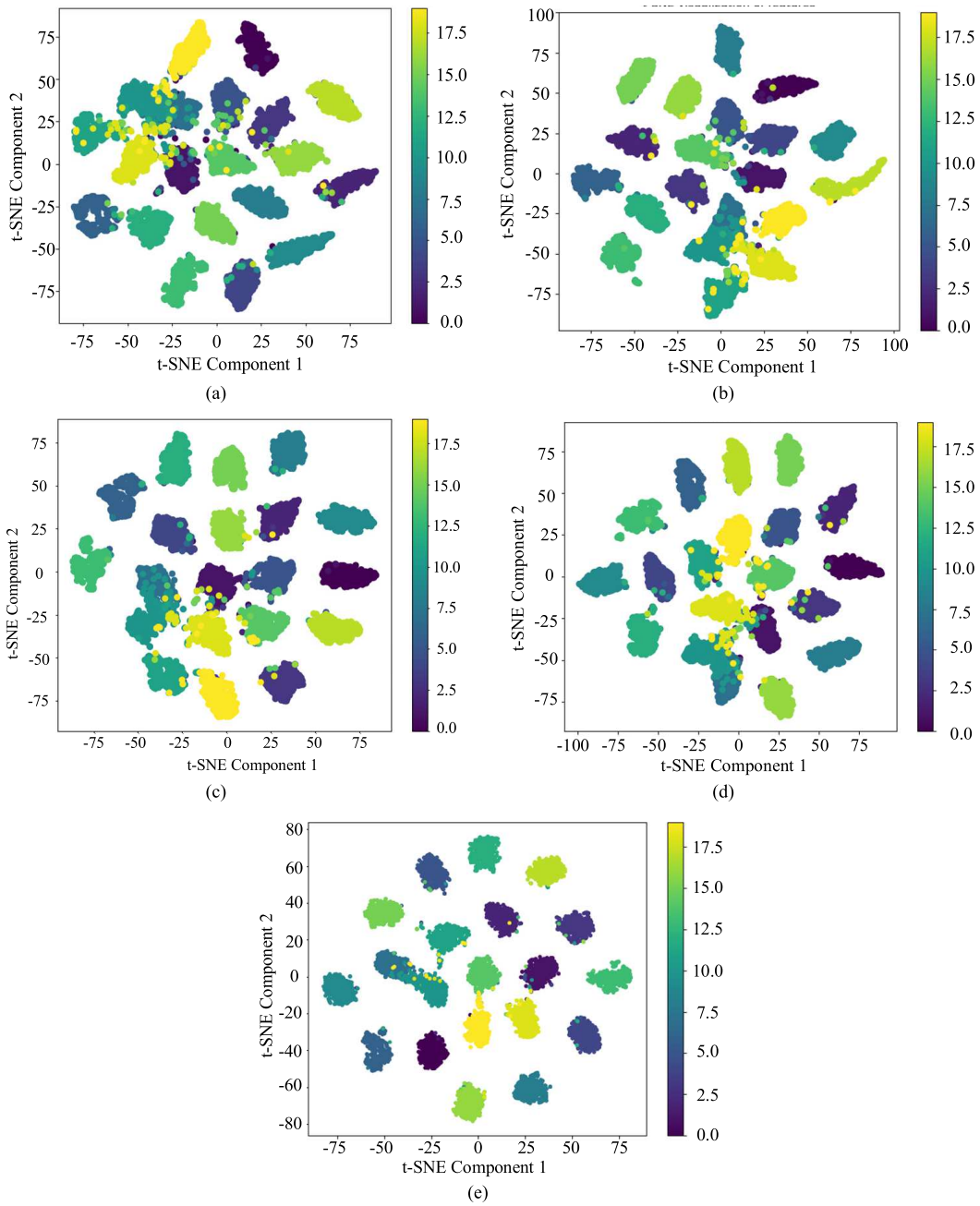


Fig. 14. Regarding the RESISC45 dataset categorization task t-SNE diagrams. (a) ResNet-50. (b) T-ResNet-50. (c) C-ResNet-50. (d) T-Chaotic ResNet-50. (e) Proposed method.

ing its general ability to avoid misidentification and maintain high recall.

Despite the higher computational cost of the proposed method compared with lightweight models (e.g., ShuffleNet CNN), it completes training and testing in 1000.8 s, which remains acceptable in practice, given its accuracy improvements. The model consumes 1.89 GB of GPU memory, falling between multifeature fusion (3.09 GB) and CFCA (0.65 GB), while achieving superior accuracy. On the RESISC45 dataset, the STMSF algorithm achieves 97.1% in accuracy, recall, $F1$ score, and precision, which demonstrates strong performance among existing remote sensing classification methods. However, a performance gap remains between STMSF and our proposed algorithm. Our method reaches 98.2% or higher across all key metrics, including accuracy, Kappa coefficient, recall, $F1$ score, and precision, clearly outperforming the 97.1% level achieved by STMSF. Furthermore, our algorithm requires only 1000.8 s for processing, which is over 75% faster than STMSF's 4116.6 s of computation time. In terms of memory usage, our algorithm consumes just 1.89 GB, representing a 77% reduction compared to STMSF's 8.37 GB, and thus offers significantly improved resource efficiency. These results demonstrate a favorable tradeoff between performance and computational cost.

The orange curve in Fig. 15 corresponds to CMTPN. Reperturbations are triggered every 20 epochs, with perturbation

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TABLE II
ALGORITHM COMPARISON EXPERIMENT OF THE RESISC45 DATASET

Algorithm	Accuracy	Kappa value	Recall rate	F1 value	Precision	Time	Space
Classifier + Detector [5]	95.4%	0.953	95.5%	95.5%	95.5%	**436.4s**	3.06G
Multi-Feature Fusion [6]	94.3%	0.940	94.3%	94.3%	94.4%	1310.0s	3.09G
GSCer [7]	94.1%	0.938	94.1%	94.1%	94.2%	1402.9s	3.25G
CFCA [8]	94.4%	0.941	94.4%	94.4%	94.4%	4062.5s	**0.65G**
STMSF [9]	97.1%	0.969	97.1%	97.1%	97.1%	4116.6s	8.37G
L2RCF [10]	97.2%	0.971	97.2%	97.2%	97.2%	1638.6s	5.33G
MGSNet [11]	95.5%	0.941	94.4%	94.4%	94.5%	1529.4s	10.01G
Improved Firefly Algorithms [12]	95.8%	0.956	95.8%	95.7%	95.8%	555.2s	1.16G
KAN [13]	95.4%	0.951	95.4%	95.4%	95.7%	685.9s	1.74G
ShuffleNet CNN [33]	96.5%	0.963	96.5%	96.5%	96.5%	483.6s	0.8G
VGG-19 [34]	90.2%	0.897	90.3%	90.3%	90.4%	1180.3s	3.11G
SnapEnsem ResNet [35]	95.9%	0.956	95.6%	95.8%	95.9%	605.1s	4.03G
Proposed algorithm	**98.2%**	**0.981**	**98.2%**	**98.2%**	**98.9%**	1000.8s	1.89G

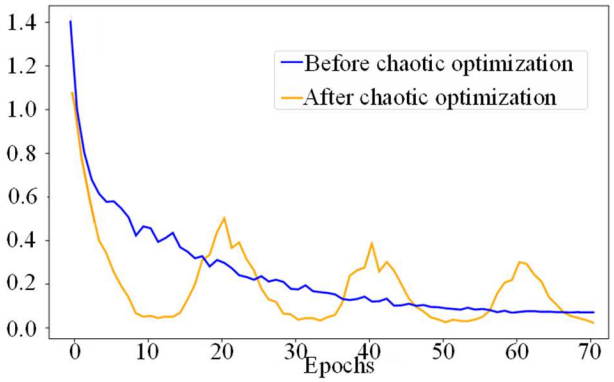


Fig. 15. Comparison chart of multiscale tensor pyramid network optimization before and after introducing chaos.

intensity governed by the chaotic map in (3). It exhibits noticeable fluctuations in the later stages of training. These fluctuations align with the re-perturbations introduced by the chaotic mechanism, which help the model to escape potential local minima or saddle points, thereby improving convergence stability.

Specifically, whenever the chaotic mapping is triggered, the model parameters undergo nonlinear updates driven by chaotic dynamics. This process often causes a brief spike or oscillation in the loss curve, which subsequently drops back to a lower level. This "fluctuation-rebound" pattern suggests that chaotic mapping enables the model to re-explore the parameter space more effectively, thereby reaching better convergence points. In contrast, the baseline ResNet (blue curve) lacks such nonlinear perturbation mechanisms. Although its loss curve generally declines, it tends to stagnate in flat or noisy regions during local phases. These findings indicate that the model incorporating chaotic mapping, through a periodic perturbation strategy, maintains the ability to escape local minima even in the later stages of training, demonstrating advantages in the final loss minimization.

B. Validation Using the AID Dataset

The dataset was randomly split using stratified sampling to preserve the original class distribution. Specifically, 50% of the samples were used for training and 50% for testing. The model was trained using the AdamW optimizer with an initial learning rate of 0.0001 and a weight decay of 8×10^{-4}. The learning rate followed a cosine annealing schedule over 70 epochs. The batch size was set to 16 for training. This difference in batch size was configured to optimize GPU memory usage and allow efficient evaluation on the entire test set within each epoch.

To mitigate class imbalance, we employed focal loss with a modulation factor $\alpha = 0.25$ and a focusing parameter $\gamma = 2.0$ [32]. The label smoothing was applied with a smoothing factor $\varepsilon = 0.1$ to enhance generalization. The training process incorporated an early stopping strategy with a patience of 40 epochs, which prevented overfitting by monitoring the validation loss. We set the parameter values of (3) to $a = 0.3, b = 0.4, c = 0.3, \mu = 1.5$, and $\lambda = 0.45$. Similar to Experiment A, to verify the effectiveness of the algorithm, we conducted two additional experiments, namely, ablation experiments and comparative experiments.

1) Ablation Experiment on the AID Dataset: The algorithms for ablation experiments include ResNet-50, T-ResNet-50, C-ResNet-50, LT-ResNet-50, TT-ResNet-50, T-Chaotic ResNet-50, and our proposed algorithm. After 20 independent experiments, the results are shown in Table III.

Table III presents the ablation study results on the AID dataset, which evaluated the performance gains from progressively introducing chaotic mapping, tensor-based representation, and multiscale pyramid strategies to the baseline ResNet-50. First, the original ResNet-50 model achieves 90.8% accuracy and 90.8% $F1$ score on this dataset, serving as a reference baseline.

The accuracy of C-ResNet-50 increases to 92.5%, indicating that chaotic dynamics help the model escape local minima and improve generalization. Meanwhile, T-ResNet-50 achieves 93.1% accuracy, showing that the tensor-based structure effectively captures richer spatial context.

Building upon this, the integration of traditional chaotic functions (e.g., logistic or tent maps) with tensor representations—denoted by LT-ResNet-50 and

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TABLE III
ABLATION EXPERIMENT OF THE AID DATASET

Algorithm	Accuracy	Kappa value	Recall rate	F1 value	Precision
ResNet-50	90.8%	0.909	90.8%	90.8%	90.8%
C-ResNet-50	92.5%	0.923	92.4%	92.4%	92.4%
T-ResNet-50	93.1%	0.930	92.9%	92.9%	92.9%
LT-ResNet-50	94.5%	0.951	95.1%	95.2%	95.2%
TT-ResNet-50	94.8%	0.950	95.0%	95.1%	95.1%
T-Chaotic ResNet-50	97.7%	0.976	97.7%	97.7%	98.2%
Proposed algorithm	**99.0%**	**0.989**	**99.0%**	**99.0%**	**99.0%**

TABLE IV
ALGORITHM COMPARISON EXPERIMENT OF THE AID DATASET

Algorithm	Accuracy	Kappa value	Recall rate	F1 value	Precision	Time	Space
Classifier + Detector [5]	94.3%	0.940	94.3%	94.3%	94.3%	**1030.3s**	3.06G
multi-feature fusion [6]	96.2%	0.960	96.2%	96.2%	96.2%	5692.5s	3.10G
GSCer [7]	93.3%	0.932	93.3%	93.3%	93.4%	5432.2s	3.25G
CFCA [8]	96.9%	0.966	96.9%	96.9%	97.0%	6688.9s	**0.65G**
STMSF [9]	98.3%	0.982	98.3%	98.3%	98.3%	4666.8s	8.37G
L2RCF [10]	98.3%	0.982	98.3%	98.3%	98.3%	4192.2s	5.33G
MGSNet [11]	97.3%	0.971	97.2%	97.2%	97.3%	2424s	10.69G
improved firefly algorithms [12]	96.1%	0.959	96.2%	96.0%	96.2%	1186.2s	1.16G
KAN [13]	95.9%	0.958	95.9%	95.9%	96.0%	2168.7s	1.74G
ShuffleNet CNN [33]	97.1%	0.969	97.1%	97.1%	97.1%	1911.4s	0.8G
VGG-19 [34]	94.9%	0.946	94.9%	94.9%	94.9%	2680.9s	3.11G
SnapEnsem ResNet [35]	95.4%	0.951	95.3%	95.4%	95.3%	1613.7s	4.03G
Proposed algorithm	**99.0%**	**0.989**	**99.0%**	**99.0%**	**99.0%**	2355s	1.89G

TT-ResNet-50—achieves 94.5% and 94.8% accuracy, respectively. These results indicate a synergistic effect between chaotic mappings and tensor representations. However, the performance gains appear to be constrained by the inherent dynamic limitations of conventional chaotic functions.

In contrast, the T-Chaotic ResNet-50, which integrates a novel chaotic mapping with tensor representations, achieves 97.7% accuracy, significantly outperforming LT-ResNet-50 and TT-ResNet-50. This result indicates that stronger chaotic dynamics and multidimensional feature extraction lead to higher discriminative power.

Finally, the CMTPN (the proposed method) achieves the highest scores on all evaluation metrics, with 99.0% accuracy, 0.989 Kappa coefficient, and both recall and $F1$ score reaching 99.0%. These results highlight that the combined use of multiscale pyramid architecture, the novel chaotic mapping, and tensor-based representation substantially boosts the model's discriminative ability and generalization performance in remote sensing scene classification.

Fig. 16 shows the t-SNE feature visualization. Fig. 16(a) shows the t-SNE visualization of the original ResNet, where the scatter points are relatively dispersed and a certain degree of category overlap is observed in the central region, indicating limited category separability. In contrast, Fig. 16(b) illustrates the feature distribution of T-ResNet, showing improved separation and reduced overlap, especially at edges. Fig. 16(c) presents the visualization of C-ResNet. Most categories form tighter clusters with clear boundaries, though scattered points persist in peripheral areas. Fig. 16(d) (T-Chaotic ResNet) further enhances clustering by combining tensors and chaos, significantly reducing point overlap, and improving intercategory separability. Finally, Fig. 16(e) demonstrates the results of the proposed method, achieving the most compact clustering and well-separated categories in both central and peripheral regions. These visualizations indicate that the proposed model achieves superior category discrimination.

2) Comparative Experiment on the AID Dataset: Table IV presents a comprehensive comparison of state-of-the-art algorithms on the AID dataset across multiple evaluation metrics, including overall accuracy, Kappa coefficient, recall, $F1$ score, precision, inference time, and GPU memory usage.

The proposed algorithm, which incorporates a novel chaotic mechanism into a multiscale tensor pyramid framework, achieves the highest performance across all metrics. Specifically, it attains an accuracy of 99.0%, a Kappa coefficient of 0.989, a recall rate of 99.0%, and an $F1$ score of 99.0%, surpassing all competing methods. In addition, it maintains a precision of 99.0%, indicating its strong ability to reduce false positives in complex multiclass scenarios.

From an efficiency perspective, although the inference time (2355 s) and memory consumption (1.89 GB) are slightly higher than lightweight models such as ShuffleNet CNN (1911.4 s, 0.8 GB), the proposed method delivers substantial accuracy gains, outperforming ShuffleNet CNN by about 2% in accuracy and 2.1% in Kappa coefficient. Moreover,

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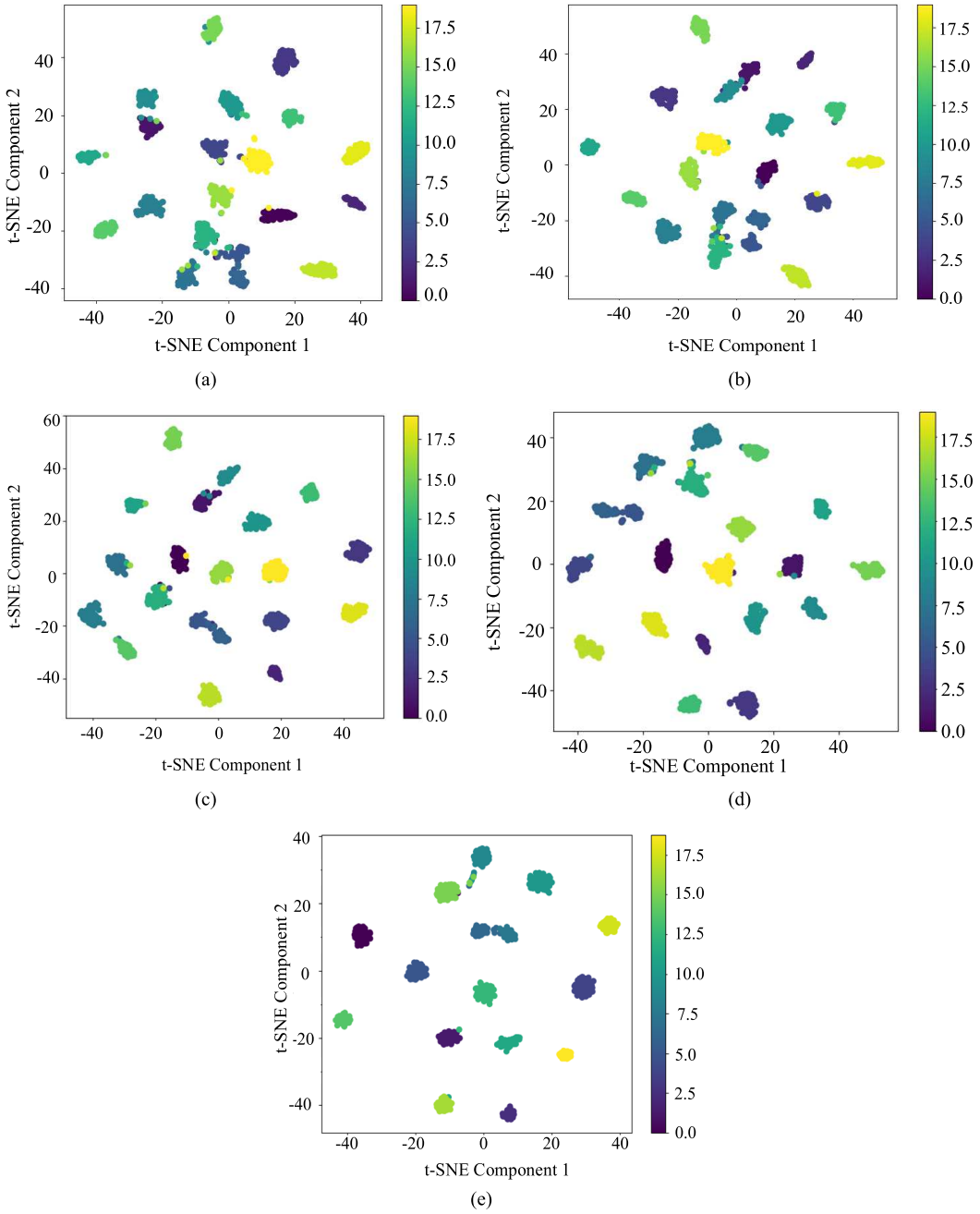


Fig. 16. Regarding the AID dataset categorization task t-SNE diagrams. (a) ResNet-50. (b) T-ResNet-50. (c) C-ResNet-50. (d) T-Chaotic ResNet-50. (e) Proposed method.

compared with computationally expensive models like multifeature fusion (5692.5 s, 3.10 GB) or GSCer (5432.2 s, 3.25 GB), the proposed approach demonstrates a more balanced tradeoff between performance and resource usage. The STMSF algorithm demonstrates strong performance on the AID dataset, with accuracy, recall, and $F1$ scores all reaching 98.3%, which represents a significant improvement over earlier methods. However, compared with our proposed algorithm, STMSF still shows slight gaps across several metrics. Our method achieves 99.0% or higher in accuracy, recall, $F1$ score, Kappa coefficient, and precision, outperforming STMSF's corresponding range of 98.2%–98.3%. Furthermore, our algorithm requires only 2355 s and 1.89 GB of memory, in contrast to STMSF's 4666.8 s and 8.37 GB. These results indicate that, while achieving higher classification accuracy, our method also offers higher computational efficiency and lower memory usage, making it particularly suitable for practical applications in remote sensing scene classification, where both accuracy and efficiency are essential.

These results confirm that the proposed method not only enhances classification accuracy and robustness but also maintains acceptable computational efficiency, making it highly suitable for real-world remote sensing scene classification tasks that demand both precision and scalability.

C. Validation Using the OPTIMAL-31 Dataset

The dataset was randomly split using stratified sampling to preserve the original class distribution. Specifically, 50% of the

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TABLE V
ABLATION EXPERIMENT OF THE OPTIMAL-31 DATASET

Algorithm	Accuracy	Kappa value	Recall rate	F1 value	Precision
ResNet-50	89.5%	0.892	89.5%	89.5%	90.7%
C-ResNet-50	90.9%	0.906	90.9%	90.7%	91.9%
T-ResNet-50	90.0%	0.897	90.1%	90.1%	91.6%
LT-ResNet-50	92.0%	0.917	91.9%	91.8%	92.8%
TT-ResNet-50	92.7%	0.925	92.7%	92.8%	93.3%
T-Chaotic ResNet-50	93.3%	0.931	93.3%	93.1%	93.7%
Proposed algorithm	**96.0%**	**0.958**	**96.0%**	**96.0%**	**96.3%**

samples were used for training, 30% for validation, and 20% for testing. The model was trained using the AdamW optimizer with an initial learning rate of 0.0001 and a weight decay of 8×10^{-4}. A cosine annealing schedule was employed to adjust the learning rate over a total of 70 training epochs. The batch size was set to 16 for training and validation and 32 for testing. This batch configuration ensured stable convergence and efficient GPU memory utilization during evaluation. The focal loss was used to address class imbalance, with the modulation factor $\alpha = 0.25$ and focusing parameter $\gamma = 2.0$. The label smoothing was applied with a smoothing factor $\varepsilon = 0.1$ to prevent overfitting and improve generalization [32]. Early stopping was implemented with a patience of 40 epochs based on the validation loss. We set the parameter values of (3) to $a = 0.3$, $b = 0.4$, $c = 0.3$, $\mu = 1.5$, and $\lambda = 0.45$. Similar to Experiment A, to verify the effectiveness of the algorithm, we conducted two additional experiments, namely, ablation experiments and comparative experiments.

1) Ablation Experiment on the OPTIMAL-31 Dataset: The algorithms for ablation experiments include ResNet-50, T-ResNet-50, C-ResNet-50, LT-ResNet-50, TT-ResNet-50, T-Chaotic ResNet-50, and our proposed algorithm. After 20 independent statistical experiments, the results are shown in Table V.

From Table V, the proposed algorithm achieves the best performance across all metrics. By integrating a multiscale tensor pyramid structure, our method effectively enhances feature representation capacity. Furthermore, the integration of a novel chaotic mechanism optimizes training dynamics, thereby improving generalization ability. Specifically, T-ResNet-50 (90.0%) marginally outperforms the baseline ResNet-50 (89.5%), suggesting that the tensor mechanism contributed to feature extraction, albeit with limited gains. In contrast, T-Chaotic ResNet-50 (93.3%) exhibits a significant improvement over T-ResNet-50 (90.0%), indicating that chaotic mapping synergizes with the tensor mechanism to optimize feature learning. Finally, our proposed algorithm attains 96.0% accuracy, substantially surpassing T-Chaotic ResNet-50 (93.3%). This demonstrates that the multiscale tensor pyramid, built upon the T-Chaotic framework, further refines the hierarchical feature representation.

Fig. 17 shows the t-SNE feature visualization maps. The t-SNE visualization results of the five models reveal that the original ResNet exhibits a relatively scattered feature distribution, with significant interclass overlap and indistinct boundaries. Introducing either the tensor mechanism or the chaotic mechanism individually leads to improvements in edge-class clustering and separation within the central region. Combining both mechanisms further enhances interclass separability and extends feature coverage, although some directional skewness remains. Building on these foundations, the proposed multiscale tensor pyramid network substantially improves spatial structural modeling and interclass discrimination. It achieves the most compact category clusters, the clearest boundaries, and the most balanced distribution of features, thereby demonstrating superior discriminative capacity and generalization in high-dimensional feature space.

2) Comparative Experiment on the OPTIMAL-31 Dataset: We further compared the proposed algorithm with several state-of-the-art remote sensing scene classification methods. The comparative experiments include classifier + detector [5], multifeature fusion [6], GSCer [7], CFCA [8], STMSF [9], L2RCF [10], MGSNet [11], improved firefly algorithms [12], KAN [13], ShuffleNet CNN [33], VGG-19 [34], SnapEnsemResNet [35], and our proposed algorithm. After 20 independent statistical experiments, the results are shown in Table VI.

The results demonstrate that our approach achieves a desirable balance among classification accuracy, robustness, and computational efficiency. It not only outperforms existing methods in mainstream classification metrics but also exhibits strong practicality with respect to both time and memory consumption.

The improved firefly algorithm is a metaheuristic optimization method inspired by swarm intelligence. It performs the search in the solution space by simulating the attraction mechanism based on light intensity among individuals. Although this method generally performs well, its optimization ability is limited by a fixed global–local search strategy and the lack of a dynamic perturbation mechanism. In contrast, the CMTPN proposed in this study introduces a nonlinear chaotic mapping, which dynamically perturbs the parameter space during training. This mechanism effectively helps the model to escape from local optima that are difficult for traditional algorithms to overcome, thereby demonstrating enhanced robustness and convergence stability in complex or noisy environments.

The attention mechanism in KAN remains relatively static after training, which limits its adaptability to shifts in feature distribution or noisy environments. In contrast, the chaotic mechanism proposed in our framework introduces

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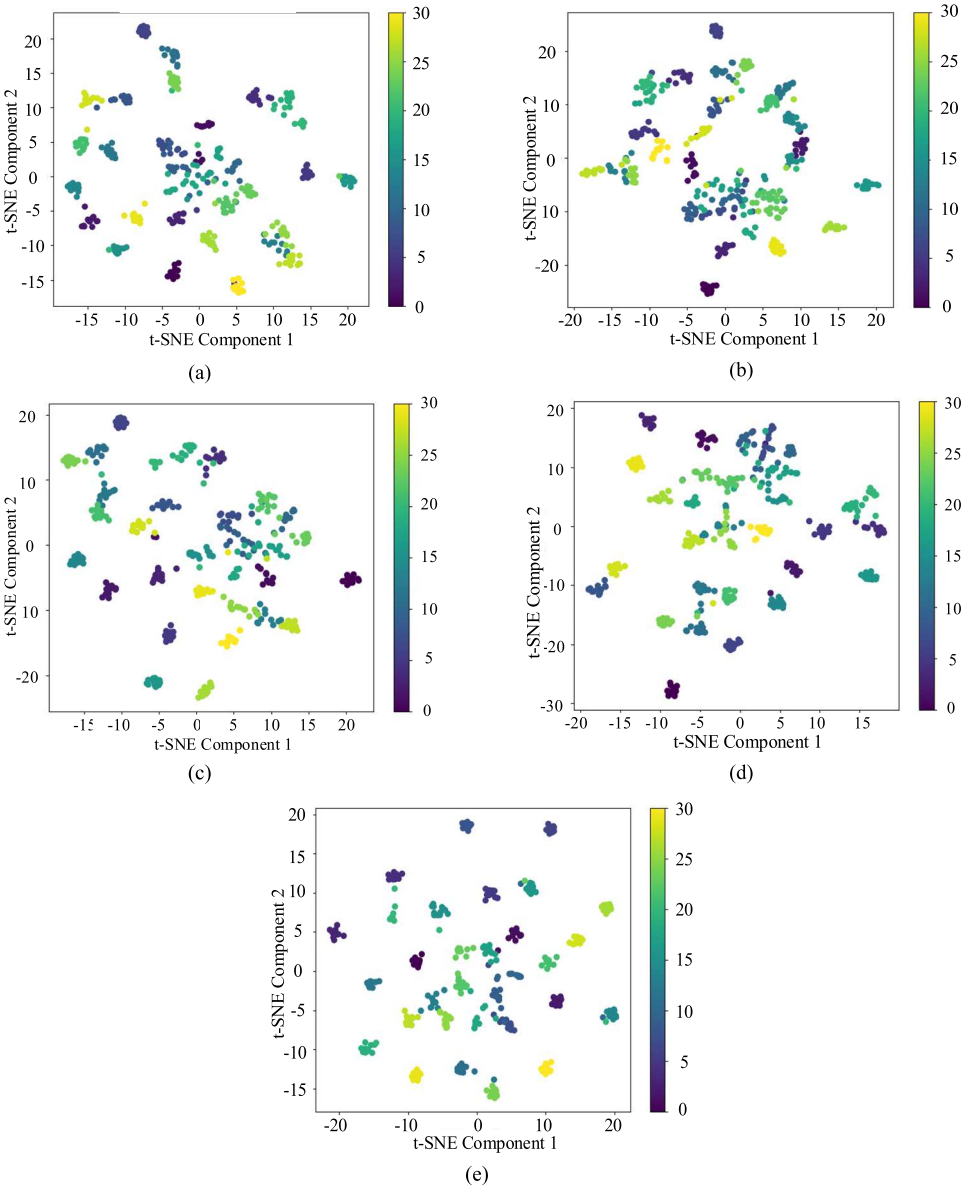


Fig. 17. Regarding the OPTIMAL-31 dataset categorization task t-SNE diagrams. (a) ResNet-50. (b) T-ResNet-50. (c) C-ResNet-50. (d) T-Chaotic ResNet-50. (e) Proposed method.

controlled nonlinear dynamics and periodic parameter perturbations. These components allow the model to escape local minima more effectively and explore a broader parameter space. Such dynamic adaptability significantly enhances robustness.

Compared with existing state-of-the-art methods, the proposed CMTPN demonstrates superior robustness and generalization across diverse datasets and evaluation metrics. It consistently outperforms other algorithms in terms of accuracy, Kappa coefficient, recall, and $F1$ score, while maintaining acceptable computational efficiency (inference time and GPU memory usage). These results validate that the integration of chaotic dynamics (Section II-A) and tensor-based multiscale modeling (Section II-B) not only enhances discriminative power but also improves model stability. Such improvements enable the framework to adapt effectively to complex and diverse remote sensing scenarios.

To evaluate the robustness and generalization of the proposed model, we conducted experiments on the above three remote sensing scene classification benchmarks. These datasets vary significantly in scene types, image resolutions, and acquisition sources, thus naturally forming a cross-domain evaluation setting. Unlike domain adaptation tasks that explicitly shift between labeled and unlabeled distributions, our "natural cross-domain" setting refers to assessing model performance consistency across structurally and geographically distinct datasets under supervised settings. The scoring results are shown in Table VII.

Each experiment involves seven evaluation metrics, including five accuracy-oriented scores (accuracy, Kappa, recall,

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TABLE VI
ALGORITHM COMPARISON EXPERIMENT OF OPTIMAL-31 DATASET

Algorithm	Accuracy	Kappa value	Recall rate	F1 value	Precision	Time	Space
Classifier + Detector [5]	90.9%	0.900	90.9%	90.9%	91.8%	**209.5s**	3.06G
Multi-Feature Fusion [6]	91.3%	0.900	90.3%	90.3%	91.3%	515.9s	3.09G
GSCer [7]	87.1%	0.867	87.1%	86.8%	88.0%	548.2s	3.25G
CFCA [8]	90.1%	0.897	90.1%	90.0%	91.3%	2850.0 s	**0.65G**
STMSF [9]	93.8%	0.936	93.8%	93.9%	94.5%	1330.8s	8.37G
L2RCF [10]	94.0%	0.939	94.0%	94.0%	94.2%	538.2s	5.33G
MGSNET [11]	90.1%	0.897	90.1%	90.1%	91.4%	506.4s	10.01G
Improved Firefly Algorithms [12]	91.1%	0.908	91.1%	91.0%	91.8%	229.7s	1.16G
KAN [13]	88.2%	0.878	88.2%	88.0%	89.9%	243.8s	1.74G
ShuffleNet CNN [33]	92.2%	0.917	91.9%	91.9%	92.5%	218.7s	0.8G
VGG-19 [34]	86.7%	0.861	86.6%	86.2%	87.9%	455.0s	3.11G
SnapEnsem ResNet [35]	89.0%	0.886	89.0%	88.8%	90.0%	245.1s	4.03G
Proposed algorithm	**96.0%**	**0.958**	**96.0%**	**96.0%**	**96.3%**	500.4s	1.9G

TABLE VII
AVERAGE NONPARAMETRIC RANKINGS OF ALL COMPARED METHODS ACROSS THREE REMOTE SENSING DATASETS

Algorithm	Average Rank
Classifier + Detector [5]	7.12
Multi-Feature Fusion [6]	8.30
GSCer [7]	11.57
CFCA [8]	7.76
STMSF [9]	5.36
L2RCF [10]	4.60
MGSNET [11]	8.05
Improved Firefly Algorithms [12]	5.29
KAN [13]	7.88
ShuffleNet CNN [33]	3.76
VGG-19 [34]	10.95
SnapEnsem ResNet [35]	8.02
Proposed algorithm	**2.33**

$F1$ score, and precision) and two complexity-related ones (time complexity and memory usage). All metrics are first normalized to the [0,1] range—accuracy metrics are positively scaled, and complexity metrics are scaled inversely—to ensure directional consistency in ranking.

A nonparametric evaluation was then conducted. Specifically, rankings were computed independently within each dataset, across all metrics. We then averaged the per-dataset rankings to obtain a final cross-domain average rank for each method. This approach avoids distributional assumptions while preserving performance consistency across datasets of varying characteristics.

The proposed algorithm achieves the best average rank (2.33), followed by ShuffleNet CNN (3.76) and L2RCF (4.60), indicating strong performance across all datasets. Traditional or high-complexity models such as VGG-19 (10.95), GSCer (11.57), and MGSNet (8.05) perform poorly in the overall ranking, primarily due to either lower classification scores or high computational cost.

To further illustrate the overall performance gaps between the proposed method and competing algorithms, Fig. 18 presents a heatmap of normalized effect sizes across three datasets and seven evaluation metrics. In this heatmap, each value represents the normalized effect size of an algorithm relative to the proposed method. Negative values indicate worse performance, while positive values suggest better performance compared with the proposed approach on that metric.

The heatmap adopts a red–blue scheme, where red denotes performance closer to or better than the proposed method (effect size > 0) and blue denotes worse performance (effect size < 0). All values are scaled to the range $[-1, 1]$ to facilitate consistent visual comparison across metrics.

As shown in the figure, the proposed algorithm maintains consistent superiority over all other methods in accuracy-related metrics, where all competitors exhibit negative effect values. In contrast, models like VGG-19 and GSCer show significant deficiencies across all metrics, while more recent methods like STMSF and L2RCF perform relatively well but still fall behind in either precision or complexity dimensions. In addition, methods such as CFCA and MGSNet incur higher computational costs, particularly in time and memory usage. Overall, the heatmap confirms that the proposed approach not

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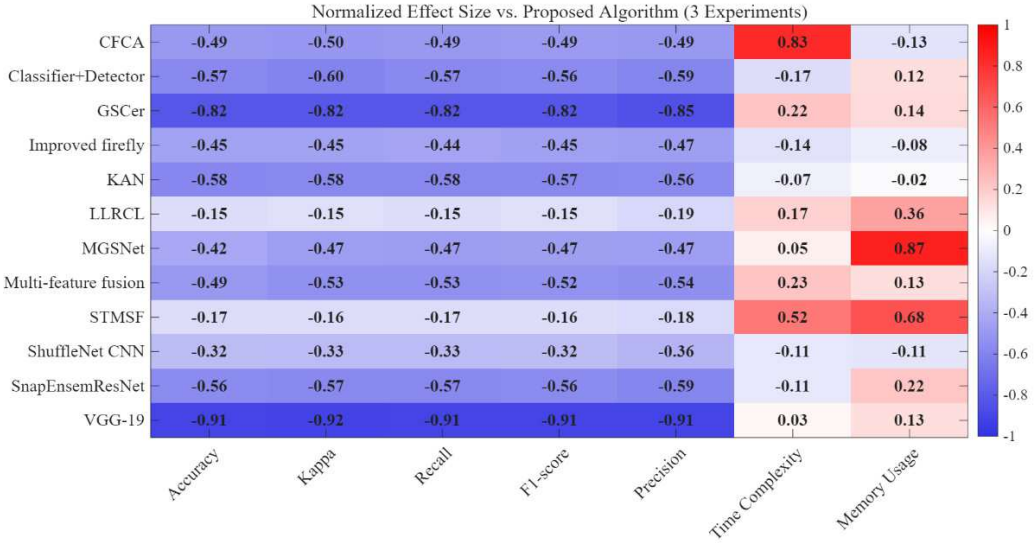


Fig. 18. Heatmap of normalized effect sizes across three experiments compared to the proposed method.

only delivers strong classification performance but also offers better efficiency tradeoffs, making it more suitable for real-world multidomain remote sensing applications.

D. Discussion

The performance of the model at different spatial resolutions is explored by comparing three datasets with different spatial resolutions—the RESISC45 dataset, the AID dataset, and the OPTIMAL-31 dataset. By analyzing the spatial resolution, computational time, and classification performance of these datasets, we are able to gain a deeper understanding of the impact of spatial resolution on remote sensing image classification tasks. The experimental results are shown in Tables II, IV, and VI. The results demonstrate that spatial resolution significantly impacts classification performance and computational complexity.

In terms of accuracy and Kappa coefficient, the proposed algorithm achieves excellent performance on all three datasets. For example, on the RESISC45 dataset, the proposed algorithm achieves 98.2% accuracy and 0.981 Kappa coefficient, which are significantly higher than the other comparison algorithms, such as the STMSF algorithm and the GSCer algorithm. Similarly, the accuracies of the proposed algorithm on the AID dataset and the OPTIMAL-31 dataset are 99.0% and 96.0%, respectively, which are also higher than those of all comparison algorithms. This indicates that high-resolution images can provide more spatial information, enabling the model to better recognize different categories in the images, thereby improving the classification accuracy.

Recall and $F1$ score are the key metrics to measure the model's performance in recognizing the correct category. The proposed algorithm also outperforms existing algorithms in these two metrics. For example, on the AID dataset, the proposed algorithm achieves an $F1$ score of 99.0%, while the STMSF and KAN algorithms achieve $F1$ scores of 98.3% and 95.9%, respectively. Similarly, it significantly improves the recall and $F1$ score on the RESISC45 and OPTIMAL-31 datasets. These results indicate that higher spatial resolution not only helps to the model recognize more details, especially in complex scenes but also significantly improves the recall, and thus, the $F1$ score.

In terms of accuracy, the proposed algorithm also demonstrates significant advantages. For example, on the RESISC45 dataset, the proposed algorithm achieves an accuracy of 98.2%, which is higher than those of other algorithms such as VGG-19 (90.2%) and SnapEnsem ResNet (95.9%). This indicates that the proposed algorithm can effectively reduce false positives (i.e., misclassification), thereby enhancing the overall classification accuracy.

Although high-resolution images usually improve performance, they also significantly increase computational time. According to the experimental results, the computational time for the AID dataset (higher resolution) is 2355 s; for the RESISC45 dataset, it is 1000.8 s; and for the OPTIMAL-31 dataset (lower resolution), it is 500.4 s. This indicates that higher resolution datasets indeed require more computational resources. Despite these challenges, the proposed algorithm's computational resource usage on high-resolution datasets remains relatively reasonable compared with other algorithms, achieving performance improvements without excessive computational overhead.

IV. Conclusion

In this study, a novel deep learning framework is proposed to improve the classification accuracy of remote sensing scene images. First, a 3-D chaotic system (Section III-A) is designed to optimize the training dynamics of deep neural networks. This system enables the model to explore a broader parameter space compared with traditional chaotic functions (i.e., logistic and tent maps), thereby accelerating convergence and obtaining a better local minimum. To further preserve multiscale spatial structural information, a tensor pyramid network architecture is employed to incorporate TCL and

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TRL. Through multilevel feature compression and cross-scale Tucker decomposition, the pyramid structure dynamically fuses multiresolution features, effectively capturing geometrical structure and high-order interchannel correlations.

Compared with existing methods based on generic CNN architectures, ResNet-50 is adopted here as the backbone network for more effective feature extraction. Ablation studies confirm the individual contributions of the proposed chaotic system and the tensor pyramid architecture. Furthermore, comparative experiments on the RESISC45, AID, and OPTIMAL-31 datasets against state-of-the-art approaches provide additional evidence for the superior performance of the proposed method.

Overall, the results indicate that the chaotic optimization reduces the risk of local minima entrapment, while the multiscale tensor pyramid enhances the spatial feature preservation, leading to the robust classification performance in complex remote sensing scenarios.

Despite the promising results, the proposed method also presents several limitations. First, the performance of the chaotic system may be sensitive to the choice of initial parameters, which could affect stability and reproducibility across datasets. Second, the incorporation of tensor operations increases the model complexity, which may lead to overfitting in scenarios with limited labeled samples or small-scale datasets.

To address these limitations, future work will focus on two directions. First, we will explore adaptive chaotic parameter tuning using principled optimization methods such as Bayesian optimization or meta-learning to improve convergence stability. Second, we plan to investigate lightweight tensor approximation strategies, including low-rank tensor decomposition and structured tensor pruning, to reduce redundancy while maintaining the representational power.

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